WSQ COURSEWARE

Agentic AI for Product Development

Lesson Plan

Course Code	**TGS-2024045799**
Technical Skills and Competencies	ICT-TEM-4035-1.1 Automation Management in Product Development
Version	**v7.0**
Release Date	11 September 2026
Training Provider	Tertiary Infotech Academy Pte. Ltd. (UEN 201200696W)
Trainer	Dr Alfred Ang

Controlled document. Verify the version before use.

Document Control

Document	Version	Date	Owner	Change summary
Lesson Plan	**6.0**	Prior release	Tertiary Infotech Academy Pte. Ltd.	Legacy chatbot courseware retained as historical reference
Lesson Plan	**7.0**	11 September 2026	Tertiary Infotech Academy Pte. Ltd.	Agentic AI product-development remap using both supplied eBooks; visual deck, twelve independent activities, aligned WA-SAQ and PP

Contents

1. Course and Delivery Information

Course	Agentic AI for Product Development
Course code	TGS-2024045799
TSC	ICT-TEM-4035-1.1 Automation Management in Product Development
Duration	2 days / 16 training hours
Assessment	2 hours: Written Assessment 60 minutes + Practical Performance 60 minutes
Delivery mode	Facilitated classroom or synchronous learning with an approved agentic AI builder
Required resources	207-slide deck, Learner Guide, twelve activity folders, spreadsheet/JSON viewer and an approved agentic AI builder

2. Learning Outcomes

- LO1 Evaluate agentic AI builders for product research and development, comparing strengths, limitations, resources and skills.
- LO2 Apply optimisation techniques to improve the efficiency, quality and control of agentic product-development workflows.
- LO3 Evaluate the benefits and trade-offs of deploying agentic AI-powered product solutions and recommend a governed path forward.

3. Facilitation Principles

- Use the PPT for mechanism visualisation, technical dialogue and evidence interpretation; do not demonstrate click-by-click procedures from the slides.
- Use the Learner Guide and activity PDFs for detailed procedures, prompts, troubleshooting and acceptance checks.
- Keep live customer data, external publishing and irreversible actions outside learner scope unless the trainer stages a safe simulation.
- Require observable execution evidence before accepting an agent response or learner conclusion.

4. Day 1 Detailed Schedule

Time	Topic / activity	Method	Resources	Evidence / outcome
09:00–09:30	Administration, attendance, context, LMS access and outcomes	Facilitated briefing	Slides 1–16	Attendance confirmation; learner context and materials access
09:30–11:00	Topic 1: product opportunity, JTBD and research-agent evidence	Mechanism dialogue + worked examples	Slides 17–40	Opportunity brief and claim ledger
11:00–13:00	Topic 1: builder comparison, readiness and build-buy-partner decisions	Technical explanation + critique	Slides 41–77	Weighted matrix and readiness heatmap
13:00–14:00	Lunch	—	—	—
14:00–15:30	Activities 01–03:	Guided activity; individual	Activity 01–03 folders +	Opportunity brief,

Time	Topic / activity	Method	Resources	Evidence / outcome
	opportunity, builder comparison and evidence pack	evidence	LG §6.1–6.3	comparison and claim ledger
15:30–17:15	Topic 2: product context, agent roles, handoffs and evaluation sets	Mechanism dialogue + technical demonstration	Slides 78–113	Workflow canvas and evaluation contract
17:15–18:00	Activities 04–06: persona, workflow canvas and AI-ready PRD	Guided setup + facilitated debrief	Activity 04–06 folders + Topic 2 checkpoint	Persona/JTBD map, handoff contract and PRD

5. Day 2 Detailed Schedule

Time	Topic / activity	Method	Resources	Evidence / outcome
09:00–10:00	Activities 07–08: prompt evaluation set and prototype experiment	Guided activity; verifier-led evidence	Activity 07–08 folders	Frozen evaluation set and experiment ledger
10:00–11:30	Topic 3: deployment posture, architecture, security and observability	Mechanism dialogue + technical demonstration	Slides 139–169	Deployment decision and control rationale
11:30–13:00	Activities 09–10: deployment and risk-control review	Guided activity; individual verification	Activity 09–10 folders + LG §6.9–6.10	Reference architecture and risk register
13:00–14:00	Lunch	—	—	—
14:00–15:00	Topic 3: KPI feedback, rollout and risk-adjusted recommendation	Evidence analysis + worked comparison	Slides 170–199	Monitoring scorecard and rollout gate
15:00–16:00	Activities 11–12: KPI dashboard and executive recommendation	Individual capstone + decision review	Activity 11–12 folders + LG §6.11–6.12	KPI tree and bounded recommendation
16:00–17:00	Written Assessment (SAQ)	Individual assessment	Assessment paper	Six responses covering K1–K6
17:00–18:00	Practical Performance	Individual performance assessment	Assessment tasks + activity evidence	Three tasks covering A1–A3

6. Assessment and Evidence Plan

Component	Duration	Coverage	Evidence	Assessor control
Written Assessment (SAQ)	60 min	K1–K6	Six open-ended responses	Identity check, independent completion, complete answer capture
Practical Performance	60 min	A1–A3	Builder evaluation, optimised workflow and governed deployment recommendation	Observe task performance; preserve learner evidence; record outcome

7. Trainer Preparation Checklist

- Confirm the course identity, version and 207-slide deck before class.
- Verify that every activity folder contains README/PDF, brief.yaml, prompt, mock data, solution, verifier and evidence checklist.

- Use a learner-safe agentic AI environment; keep live data, unapproved integrations and external publishing out of scope.
- Confirm assessment candidate papers are learner-accessible; retain answer keys in trainer-private storage only.
- Test LMS and attendance access without exposing credentials on projected screens.

8. References

- **Current course page:** https://www.tertiarycourses.com.sg/wsq-agentic-ai-for-product-development.html — Course identity, duration, learning outcomes, topics and assessment format
- **AI Product Management:** Local reference/AI Product Management.pdf — AI-PM playground, natural-language development, rapid prototyping, specification, iteration and production handoff
- **Building AI-Powered Products:** Local reference/Building AI-Powered Products.pdf — AI product-development lifecycle, product sense, goals and metrics, agent patterns, evaluation and production rollout
- **Building effective agents:** https://www.anthropic.com/engineering/building-effective-agents — Agent/workflow distinction, simple composable patterns and complexity trade-offs
- **People + AI Guidebook:** https://pair.withgoogle.com/guidebook-v2/ — User needs, success definition, mental models, feedback, control and graceful failure
- **NIST AI RMF:** https://www.nist.gov/itl/ai-risk-management-framework — Govern, map, measure and manage risk throughout the AI lifecycle